EXPLAINABLE AI

PHISHING EMAIL DETECTION WITH LIME

A DistilRoBERTa classifier predicts phishing. LIME explains the local evidence behind one email at a time.

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STUDY OVERVIEW

Problem: binary email classification
`1` phishing · `0` safe

Need: security analysts need reasons, not only a model score.

Approach: train a transformer, then explain single predictions with a model-agnostic local surrogate.

DATA	SPLIT	MODEL	LIME	CHECK
clean labels	stratified	DistilRoBERTa	token weights	deletion test

DATA

18,631
USABLE EMAILS

11,322
SAFE EMAILS

7,309
PHISHING EMAILS

14,904
TRAIN ROWS

1,863
VALIDATION ROWS

1,864
TEST ROWS

The source file has 18,650 rows; 19 blank-text rows are dropped before training. Class ratio is preserved across splits.

MODEL

Fine-tuned `distilroberta-base` for one epoch using Hugging Face Transformers.

Strong test metrics are useful, but they do not explain a specific alert.

0.9667
ACCURACY

0.9579
F1-SCORE

0.9502
PRECISION

0.9658
RECALL

LIME

LIME explains one prediction by learning a simple local model around the original email.

TOKENIZE

words + punctuation

MASK

remove tokens randomly

SCORE

ask transformer

WEIGHT

near samples higher

FIT

ridge regression

Positive weight means the token increases phishing probability.

Negative weight means the token decreases phishing probability in this local explanation.

EXAMPLE

> Your account has been suspended. Click here to verify.

0.9655

PREDICTED PHISHING PROBABILITY

Phishing tokens: Your · Click · here · verify

Safe-side local tokens: has · account · been

A token's sign is local. It is not a global dictionary of safe or malicious words.

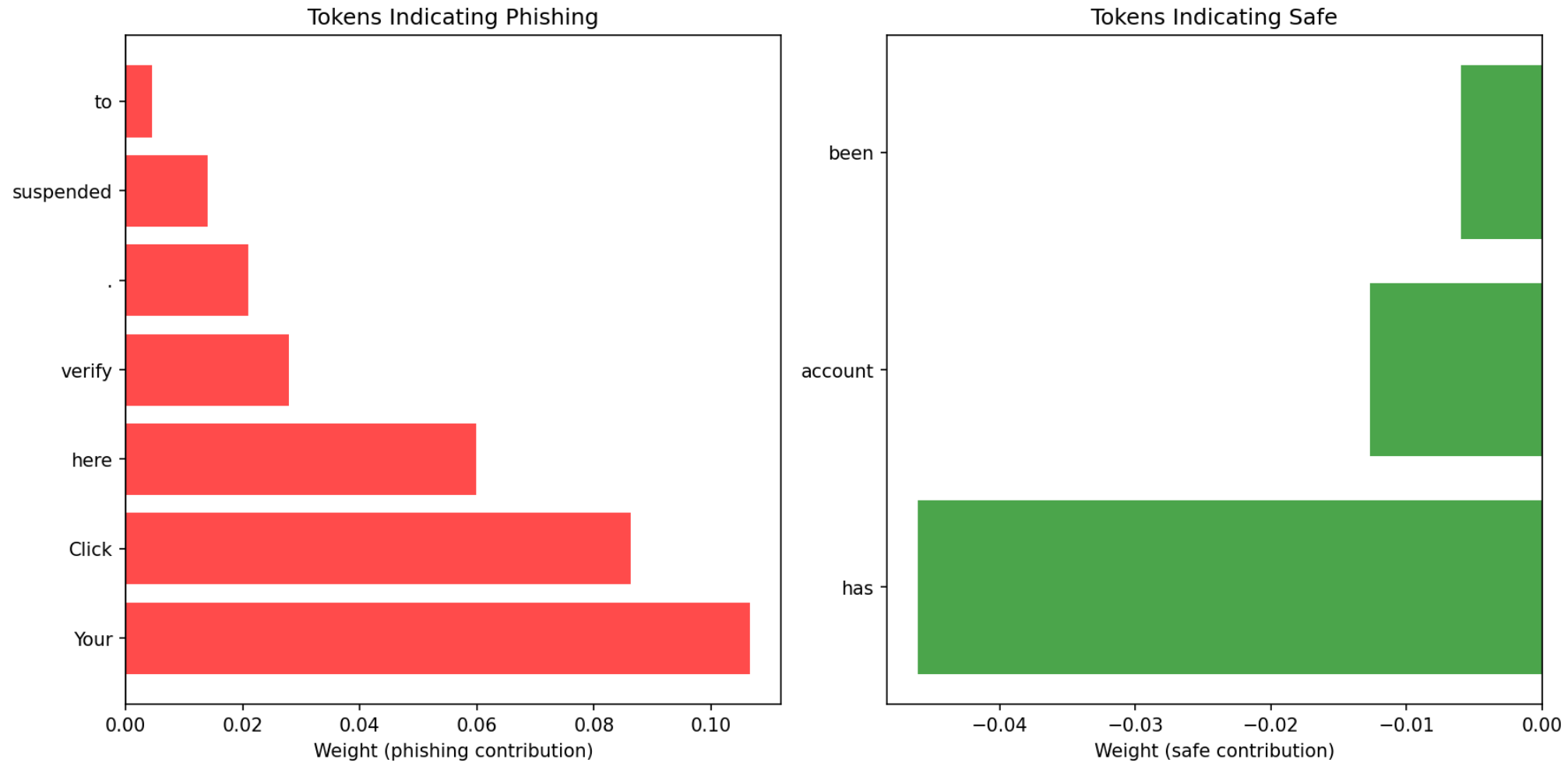
LIME OUTPUT

```
==================================================
EXPLANATION SUMMARY
==================================================
Prediction: PHISHING
Probability: 0.9655

Top phishing-indicating tokens:
  + Your            (0.1066)
  + Click           (0.0862)
  + here            (0.0599)
  + verify          (0.0279)
  + .               (0.0209)

Top safe-indicating tokens:
  - has             (-0.0460)
  - account         (-0.0127)
  - been            (-0.0060)
==================================================
```

TOKEN IMPORTANCE



DELETION TEST

Remove high-importance tokens first and compare the phishing score against random deletion.

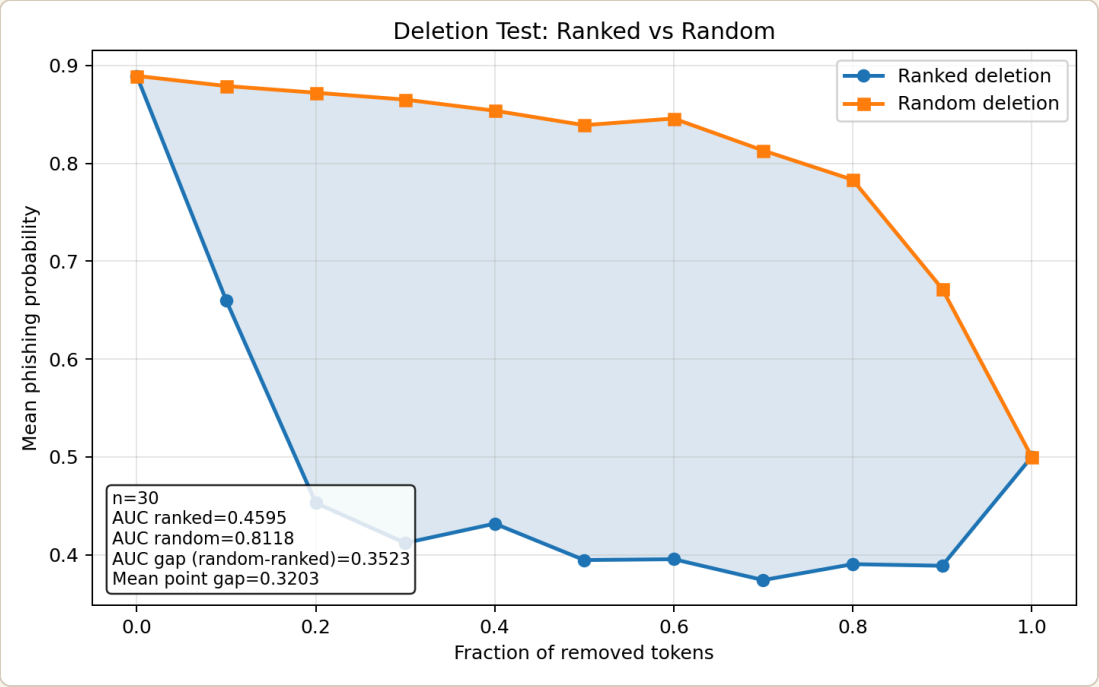
0.4595

RANKED AUC

0.8118

RANDOM AUC

Lower AUC is better. The ranked curve drops faster over 30 phishing emails.



LIMITATIONS

- Word removal can create unnatural emails.
- Correlated phishing phrases can split credit.
- LIME explains one prediction, not the whole model.

Useful next step: run the deletion test directly with LIME-ranked tokens.